

Master's Comprehensive Examination

Part I

Theory (90 minutes)

December 7, 1996

Instructions: Answer all four questions. Open book and open notes.

1. (a) Draw 2 chips from 5 chips numbered 1 through 5 without replacement. Let Y be the sum of the chips. Find EY and $\text{Var } Y$.

(b) Find the probability that the second chip drawn is at least 4.

2. Let X be a random variable whose density is constant on the interval $(1, 2)$ and whose density $f_X(x) = x$ for $0 < x < 1$.

(a) If C denotes the constant, find C .

(b) Find $P\{1/2 < X < 3/2\}$.

(c) Find the c.d.f. of X .

3. Let X_1, X_2 be a random sample of size 2 from a gamma distribution with parameters $\alpha = 3$ and θ . That is

$$f_X(x; \theta) = \frac{e^{-x/\theta} x^2}{\Gamma(3)\theta^3}, \quad x > 0.$$

(a) Find a sufficient statistic T .

(b) Is $\log_e(X_1 + X_2)$ a sufficient statistic?

4. Let X_1, X_2 be a random sample of size 2 from a Poisson distribution with parameter λ . Test $H_0 : \lambda = 1$ vs $H_1 : \lambda > 1$. Find the UMP test of size $\alpha = .0526$. (Note that $(X_1 + X_2) \sim P(2\lambda)$.)

Master's Comprehensive Examination

Part II

Applied (90 minutes)

December 7, 1996

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1. A simple linear regression model

$$Y = \beta_0 + \beta_1 X + \epsilon$$

was fit to a set of data. The following results were found.

$$\hat{\beta}_1 = 10$$

$$\bar{X} = 8$$

$$\bar{Y} = 2$$

$$n = 16$$

In addition, part of the analysis of variance table is summarized below:

Source	dF	SS	MS	F
Regression		1000		
Error			100	
Total				

- (a) Find the estimated regression model.
- (b) Using the ANOVA table, test

$$H_0 : \beta_1 = 0 \quad (\text{let } \alpha = .05).$$

- (c) Find the 95% confidence interval for β_1 .
- (d) Find the 95% confidence interval for the mean value of Y when $X = 6$.
- (e) If the variance of ϵ is σ^2 , estimate σ^2 .
- (f) Find the coefficient of determination. What does this mean?
- (g) Estimate the correlation coefficient, ρ .

2. Please answer each of the following questions either True (T) or False (F).

- (a) The smaller the value of α (closer to 0), the higher the probability that H_0 will be accepted.
- (b) α is the probability of observing a value of the test statistic outside the acceptance region when H_0 is false.
- (c) Making α smaller (closer to 0) actually increases the probability of making a Type II error if H_0 is false.

(d) If we wish to fix the width of a confidence interval, we can obtain a confidence level $(1 - \alpha)$ as high as possible, provided we take a large enough sample.

(e) If α is increased (closer to 1), the area of the rejection region is decreased.

(f) The analysis of variance technique is so named because it is used to test a hypothesis about variances.

(g) If a 95% confidence interval contains the value of $\mu = 3$, a two-sided test $H_0 : \mu = 3$ will accept H_0 if $\alpha = 0.05$.

(h) If you are testing $H_0 : \mu \leq 0$ vs $H_a : \mu > 0$ and compute $\bar{x} = -0.361$, then H_0 will be accepted.

(i) To construct a confidence interval for the mean of a Normal distribution the t-distribution is used if the variance of the population is known.

(j) The mean of Student's t-distribution is always 0.

3. A nested factorial experimental design was used by a researcher. Factor A , which was a fixed effect, was a between subject factor. Subject, a random effect, was nested within factor A . Each subject was evaluated at each level of factor B , which was a fixed effect.

The experiment was balanced, meaning the same number of subjects were assigned to each level of A and there were no missing data. Below is a partially completed analysis of variance table.

Source	df	SS	MS	MS	F
Total	54				
A	$\square$	$\square$	$\square$	$9\sigma_{S(A)}^2 + 18\sigma_A^2$	2
$Subj(A)$	$\square$	$\square$	10	$\square$	
B	$\square$	$\square$	$\square$	$\sigma_{S(A)B}^2 + 6\sigma_B^2$	1
$A \times B$	$\square$	80	$\square$	$\square$	$\square$
$Subj(A) \times B$	$\square$	$\square$	5	$\square$	

Complete the analysis of variance table.

4. An experiment is conducted to compare two drug treatments with respect to their ability to eliminate the pain of a headache within 30 minutes. The results of the experiment are given below.

	Drug A	Drug B
Cure Yes	100	40
No	50	60

(a) Test the null hypothesis that the cure rates are equal for the two drugs using a Z-test. (Let $\alpha = .05$.)

(b) Test the null hypothesis that the cure rates are equal for the two drugs using a Chi-square test. (Let $\alpha = .05$.)

(c) Which approach (a) or (b) is correct? Why?

(d) Construct a 95% confidence interval for the cure rate for Drug A.

(e) Construct a 95% confidence interval for the difference in cure rates between Drugs A and B.

5. An experiment was conducted to determine the effectiveness of a new drug for the treatment of tapeworms in sheep. A random sample of 14 worm-infested lambs was randomly divided into two groups. Seven were injected with the new test drug and 7 were left untreated. After six months, the lambs were sacrificed and the following worm counts were recorded.

	Untreated	Treated	Untreated - Treated
	18	40	-22
	43	54	-11
	28	26	2
	50	63	-13
	16	21	-5
	32	37	-5
	13	39	-26
mean	28.57	40.00	-11.43
stand.dev.	14.09	14.67	9.91

(a) The objective of this experiment was to determine if the treatment reduces the number of worms when compared to the control. Formulate the null and alternative hypothesis to meet the objectives of the experiment.

(b) Is the proper procedure a paired or an unpaired t-test? Justify your answer.

(c) Test the hypothesis you have stated in (a) using $\alpha = 0.05$.

(d) State your conclusions from part (c).

(e) Construct a 95% confidence interval for the mean difference between the treatment groups.

Master's Makeup Exam

April 23, 1996

1. (a) Assuming 365 equally likely days, determine the probability that no two persons among 6 at a party have the same birthday.
(b) A college entrance board views students as capable or incapable. The examination given to prospective students is passed by a capable student with probability .9 and with probability .3 by an incapable student. If the probability that a student taking the exam is capable is .45, what is the probability that a student who has passed is capable?
2. Suppose a card is drawn from a standard deck of 52, noted and replaced, and another is drawn and noted. If all possible pairs of draws are taken to be equally likely, what is the probability of each of the following events?
 - (a) Both cards are drawn red.
 - (b) The first card drawn is red, and the second is black.
 - (c) At least one card drawn is red.
 - (d) One card drawn is an ace, and the other is a king.
 - (e) The first card drawn is an ace, and exactly one card drawn is a diamond.
3. Let X be a Poisson variable with parameter λ ; i.e. $f_X(x; \lambda) = e^{-\lambda} \lambda^x / x!$
 - (a) Test $H_0 : \lambda = 1$ vs. $H_1 : \lambda = 2$ at level of significance .0037. I.e. indicate for what values of X you reject and what values of X you do not reject.
 - (b) Find the power of the test.
4. Let X have an exponential distribution with parameter θ ; i.e. $f_X(x; \theta) = \frac{1}{\theta} e^{-x/\theta}$. A 95% confidence interval for θ based on the pivot (X/θ) is

$$(KX, \frac{1}{-\log_e .98} X).$$

Note $e^{-(-\log_e .98)} = .98$. Find K .

Master's Comprehensive Examination

Part I

Theory (90 minutes)

March 23, 1996

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1. Let $X \sim N(\mu, 1)$ and $Y \sim N(\mu, 10)$. X and Y are independent. Let a be a constant and let $T_a = aX + (1 - a)Y$.

(i) Find ET_a and decide if it is an unbiased estimator of μ .

(ii) Find the value of a that makes the $\text{Var } T_a$ smallest.

2. Let $X_1, \dots, X_n$ be iid with $P(X_i = k) = \frac{e^{-\lambda} \lambda^k}{k!}$, where $k = 0, 1, \dots$.

(i) Find the UMVUE of λ .

(ii) Find the maximum likelihood estimator (MLE) of $P(X_1 = 1)$.

3. Let X be the only observation from the density $f(x; \theta) = (\theta + 1)x^\theta I_{(0,1)}(x)$, where $\theta > -1$. In testing $H_0 : \theta \leq 1$ vs $H_1 : \theta > 1$, find the power function and size of the following test: reject H_0 iff $X > 1/2$.

4. In a vaccine study the vaccinated and non-vaccinated groups were examined for subsequent illness or not. The results are given in Table 1.

Table 1

	Sick	Well
Vaccinated	14	86
Not Vaccinated	44	56

(a) What is the probability of being vaccinated given you are sick?

(b) Are vaccine status and health status statistically independent?

(c) An intercom between Apollo and Soyuz can fail at the American end with probability .09 and at the Russian end with probability .12. The probability that both ends fail is .06. Find the probability that neither end fails.

5. (i) In rolling 12 dice, what is the probability of getting each face twice?

(ii) For a binomial random variable X with parameters $n = 4$ and $p = 1/2$, find the following conditional probability

$$P[X > 1 | X > 0].$$

Master's Comprehensive Examination

Part II

Applied (90 minutes)

March 23, 1996

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1. A simple linear regression model

$$Y = \alpha + \beta X + \epsilon$$

was fit to a set of data

$$\begin{array}{ll} n = 13 & \sum XY = 514.80 \\ \sum X = 130 & \sum X^2 = 1562 \\ \sum Y = 44.4 & \sum Y^2 = 171.30 \end{array}$$

- Find the estimated regression line.
- If it is assumed that ϵ is distributed as $N(0, \sigma^2)$, then estimate σ^2 .
- Test the hypothesis that $\beta = 0$. Let the probability of a type I error = .01.
- Find a 95% confidence interval for the mean value of Y when $X = 12$.
- Find the coefficient of determination. What does this mean?
- Find an estimate of the correlation coefficient, ρ .

2. A four factor factorial experiment has been run. The factors and the number of corresponding levels are shown below:

<i>A</i>	2 levels
<i>B</i>	3 levels
<i>C</i>	4 levels
<i>D</i>	5 levels

Using our usual notation this means that the experiment was a $2 \times 3 \times 4 \times 5$ factorial. For each treatment combination 2 observations were collected.

- How many treatment combinations does the experiment have?
- How many data points will be collected for the experiment?
- Write out the analysis of variance table including the sources of freedom, the degrees of freedom, and the expected mean squares.

3. In a certain city the ethnic distribution of its residents is:

White	65%	Hispanic	10%
Black	15%	Other	10%

The police force of this city has 500 members. Of these members their ethnic distribution is

White	250	Hispanic	40
Black	200	Other	10

Determine if the workers of the police force are a representative sample from the population.

4. The probability, p , that a given die comes up as 6 is to be tested.

$$H_0 : p = 1/6$$

$$H_a : p > 1/6$$

The die is to be rolled 600 times and X represents the number of times the die comes up 6. H_0 will be rejected if X exceeds a critical value C .

a. Find C such that the test will have a 5% level of significance.

b. For the value of C in part (a), find the power of the test against the alternative $p = 2/6$.

c. If $X = 150$, what would be the observed level of significance (p -value)?

5. A study is conducted to test a new drug in reducing diastolic blood pressure. Ten patients are evaluated over a 6 week period. Blood pressure measurements are taken at the beginning and end of the study. These data are shown below:

Patient	Pre	Post
1	105	95
2	95	100
3	100	80
4	95	90
5	100	85
6	105	85
7	90	90
8	95	85
9	90	100
10	95	90

These data are analyzed using a two-sample t -test.

a. Explain why this approach is inappropriate.

b. Carry out an appropriate test to determine if the drug is effective in reducing diastolic blood pressure. Let $\alpha = .05$.

c. Give a one-sided 90% confidence interval for the mean reduction in diastolic blood pressure.